

Data Communication and Networks (ELC 314)

Fall 2025-2026

By

Dr. Ahmed El-Eraki

Lecturer of Electronics and Communication Engineering
College of Engineering and Technology
Badr University in Cairo (BUC)

ELC 411 Course Specification

Course Outcomes :

1. Apply signal analysis to model and interpret data transmission in engineering systems.
2. Relate digital transmission concepts to system reliability through BER and channel capacity.
3. Explore analog modulation and bandwidth allocation.
4. Investigate multiplexing and spreading as efficient resource allocation methods.
5. Compare transmission media properties with transmission line concepts in power systems.
6. Analyze telephone and cable networks as case studies for legacy and modern communication.
7. Apply error detection and correction to ensure reliable data delivery, mirroring fault tolerance.
8. Examine switching protocols and architectures, relating to routing and distribution strategies.
9. Study internetworking and ISDN as integrated system models, bridging theory and applications
10. Applying software tools and simulation packages.

ELC 411 Course Specification

Course content :

1. Data and Signals
2. Digital Transmission
3. Analog Transmission
4. Bandwidth Utilization: Multiplexing and Spreading
5. Transmission Media
6. Using Telephone and Cable Networks for Data Transmission
7. Error Detection and Correction Techniques
8. Switching Protocols and Architecture
9. Integrated Services Digital Network (ISDN)

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Teaching and Learning Methods:

1. Lectures and PowerPoint Presentations
2. Discussions and In-class & Online Exercises
3. Solved Practical Examples and Self-Study
4. Homework Problems
5. Information Collection from Different Sources

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Assessment Method:

Assessment Method	Student Activity	Weight (%)	Week
Mid-Term Exam	Written Test	15%	7-8
Practical Exam	Lab/Practical Tasks	10%	14
Student Activities	Assignments/Projects	25%	All
Final Exam	Comprehensive Exam	50%	15
Total	—	100%	

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Grading Option :

Grading Option		B2		
No.	Method	Min Frequency Weekly	Weight	Frequency
1	Attendance and Participation	Weekly	2	Weekly
2	Weekly Assignments	8	8	8
3	In-class Quizzes	2	8	2
4	Lab Assignments	6	5	6
5	Project Tasks	--	--	--
6	Self-Learning Exercises (online quiz)	2	2	2
7	Max Instructor Bonus		3	
		Total	25	

ELC 411 Course Specification

Reference:

1. Handout of the of the present Lecture
2. Selected parts from textbook according to course content “Data Communications and Networking Fourth edition By-Behrouz-A.Forouzan”

Lecture 1: Introduction to Data Communication (**Ref**: Chapter 1)

Data Communication (ELC 314)

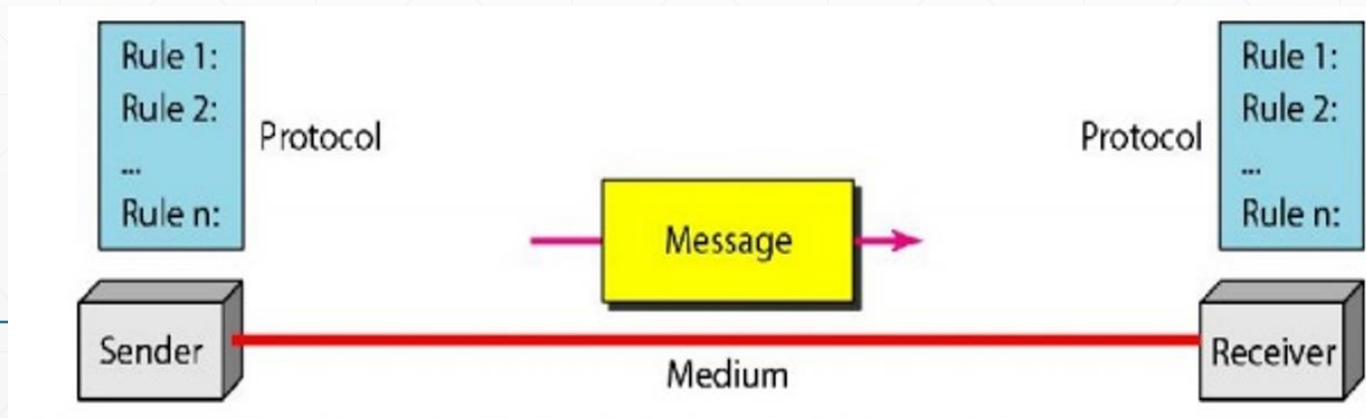
Data Communication Basics

- ❑ **Communication** : sharing of information (local face-to-face or remote via telecommunication).
- ❑ **Data** : information in an agreed form.
- ❑ **Data communication** : exchange of data between devices using a transmission medium (e.g., cable, wireless).
- ❑ Requires both **hardware** (equipment) and **software** (programs).
- ❑ Effectiveness depends on four key characteristics:
 - **Delivery** – Data must reach the correct destination.
 - **Accuracy** – Data must be delivered without errors.
 - **Timeliness** – Data must be delivered on time (real-time for audio/video).
 - **Jitter**: uneven variations in the packets' arrival times, some 10ms and others 20ms.

Data Communication System

□ A data communications system has five components:

- **Message** – The information to be transmitted (text, numbers, images, audio, video).
- **Sender** – The device that generates and sends the message (e.g., computer, phone, camera).
- **Receiver** – The device that receives the message (e.g., computer, TV, handset).
- **Transmission Medium** – The physical path for communication (e.g., twisted-pair, coaxial, fiber optics, radio waves).
- **Protocol** – A set of rules that ensure proper communication between devices.



Data Representation

❑ **Information forms:** Text, numbers, images, audio, video.

1. Text:

- ❑ Represented as bit patterns (0s and 1s).
- ❑ Each set of patterns = code (coding process).
- ❑ Unicode (32-bit): supports all world languages.
- ❑ ASCII (7-bit): early standard, first 127 Unicode characters.

2. Numbers

- ❑ Represented **directly** in **binary form**
- ❑ Facilitates mathematical operations.

3. Audio:

- ❑ Continuous by nature (sound/music).
- ❑ Captured as analog signals, later converted to digital or analog formats for transmission.

4. Images

- ❑ Composed of **pixels** arranged in a matrix.
- ❑ Higher pixel count = better resolution, but more memory.
- ❑ Each pixel assigned a bit pattern:
- ❑ **1-bit:** black/white.
- ❑ **Multi-bit:** grayscale or color.
- ❑ **Color models:**
 - **RGB** (Red, Green, Blue).
 - **YCM** (Yellow, Cyan, Magenta).

5. Video:

- ❑ Sequence of images (frames) to represent motion.
- ❑ Can be continuous (TV camera) or discrete (frame-based).
- ❑ Converted into digital or analog signals for communication.

Data Flow (Transmission Modes)

1.Simplex

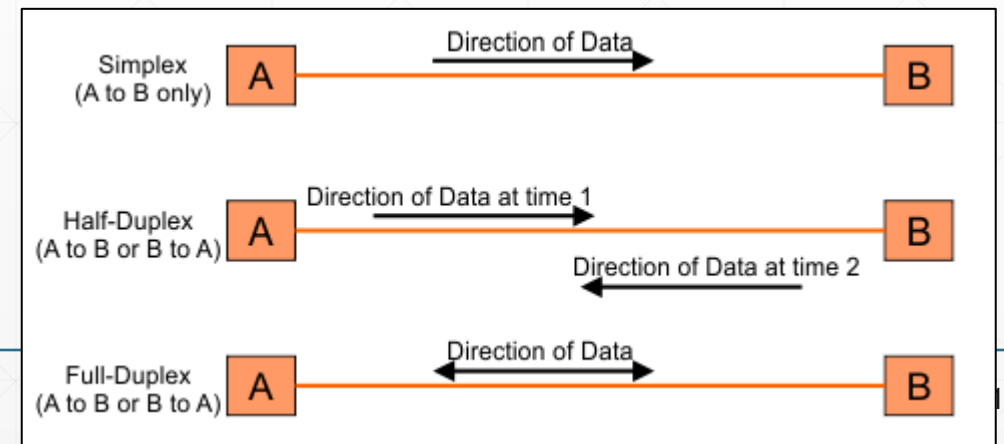
- **One-way communication only** (like a one-way street).
- One device transmits, the other only receives.
- **Examples:** Keyboard → Computer, Monitor ← Computer.
- Entire channel capacity used in one direction.

2.Half-Duplex

- **Two-way communication, but not simultaneous** (like a single-lane road).
- At any moment, only one device can transmit while the other receives.
- **Examples:** Walkie-talkies.
- Channel capacity fully used in one direction at a time.

3.Full-Duplex

- **Two-way communication simultaneously** (like a two-way street).
- Signals in both directions share the channel capacity.
- Achieved by:
 - Two separate transmission paths, or
 - Splitting channel capacity.
- **Example:** Telephone conversation.



Networks

- A network is a set of devices (often referred to as *nodes*) connected by communication links.
- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Networks Criteria

Criteria	Definition	Key Metrics / Factors
Performance	Ability of the network to deliver data efficiently.	- Transit time (delivery speed) - Response time (inquiry → reply) - Throughput (data transfer rate) - Delay (time taken) - Factors: users, medium, hardware, software
Reliability	Accuracy of data delivery over time.	- Frequency of failures - Recovery time after failure - Robustness during critical events
Security	Protection of data and systems against threats.	- Prevent unauthorized access - Prevent data damage/alteration - Data recovery policies & procedures

Physical Structures

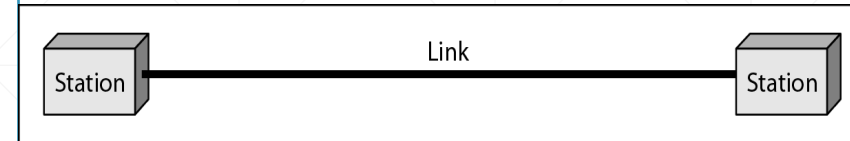
1.Type of Connection

2.Physical Topology

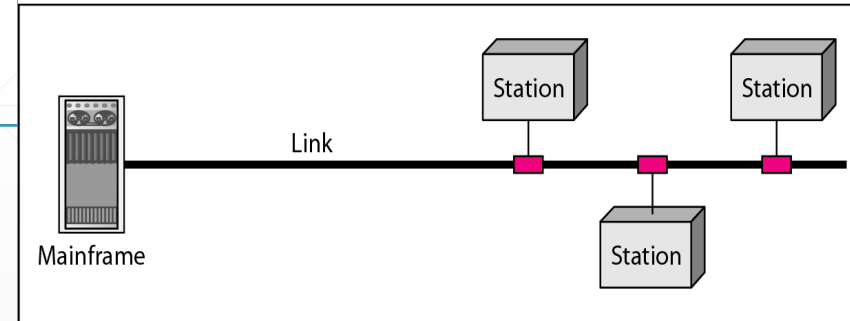
Physical Structures

Type of Connection

Aspect	Types / Details	Advantages	Disadvantages
Point-to-Point	Dedicated link between two devices (e.g., TV remote ↔ TV)	full capacity for two devices	Point-to-point: costly with many devices
Multipoint (Multidrop)	Single link shared by multiple devices (spatially or time-shared)	efficient use of shared medium	performance drops if overloaded



a. Point-to-point

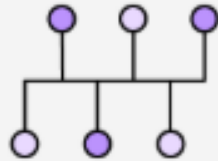


b. Multipoint

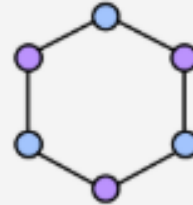
Physical Structures

Physical Topology

Bus



Ring



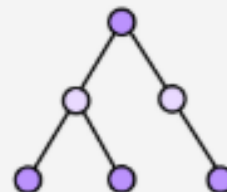
Mesh



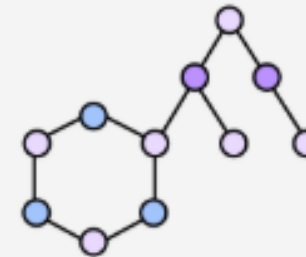
Star



Tree

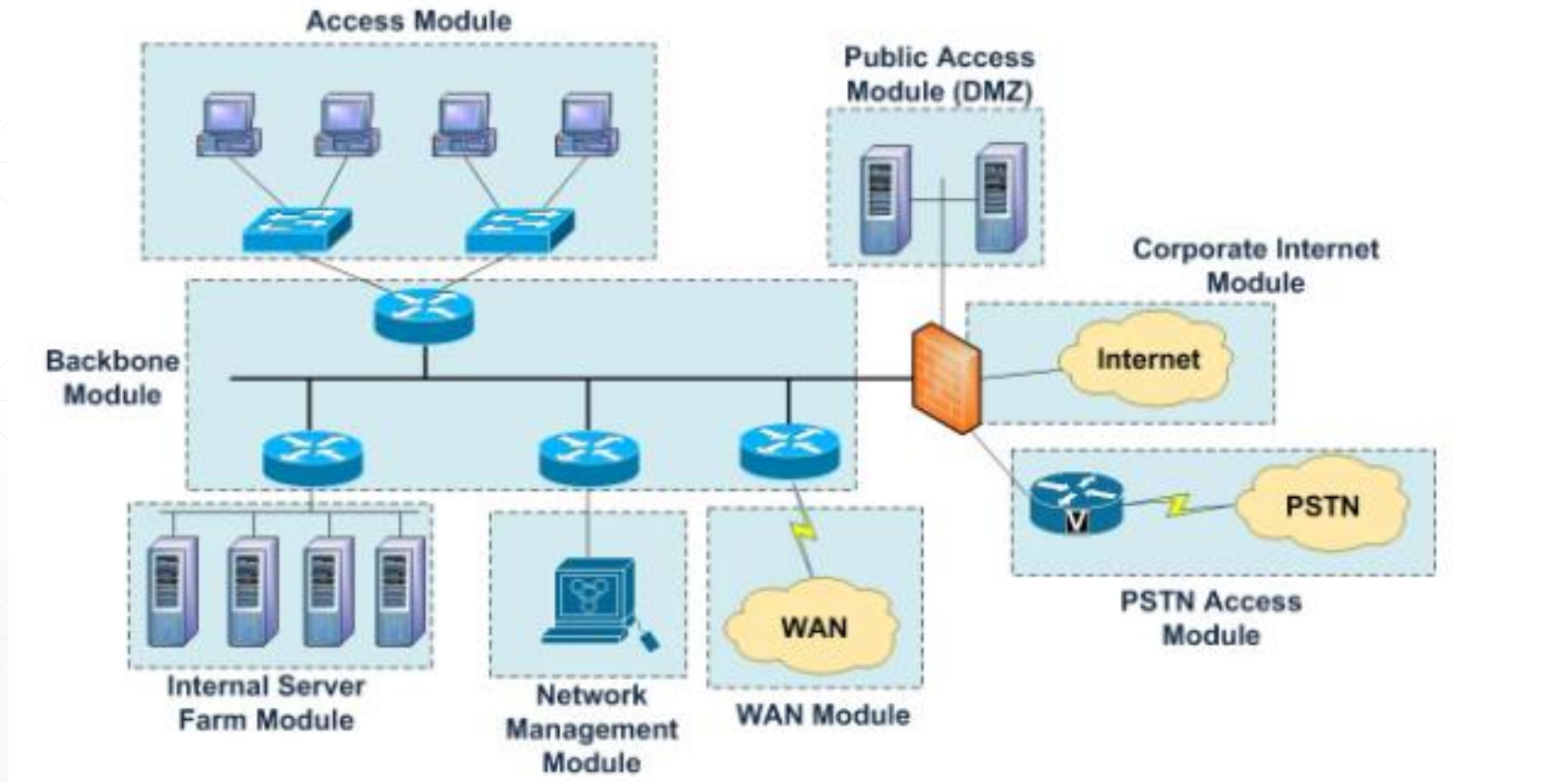


Hybrid



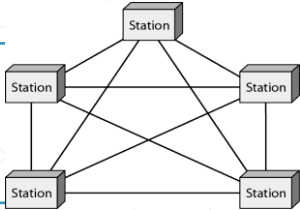
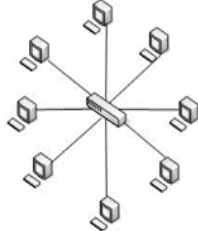
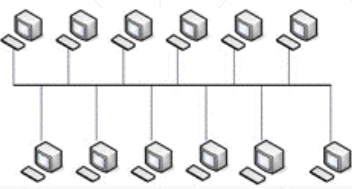
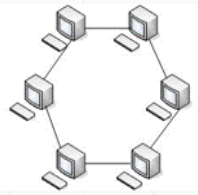
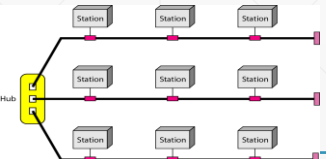
Physical Structures

Physical Topology

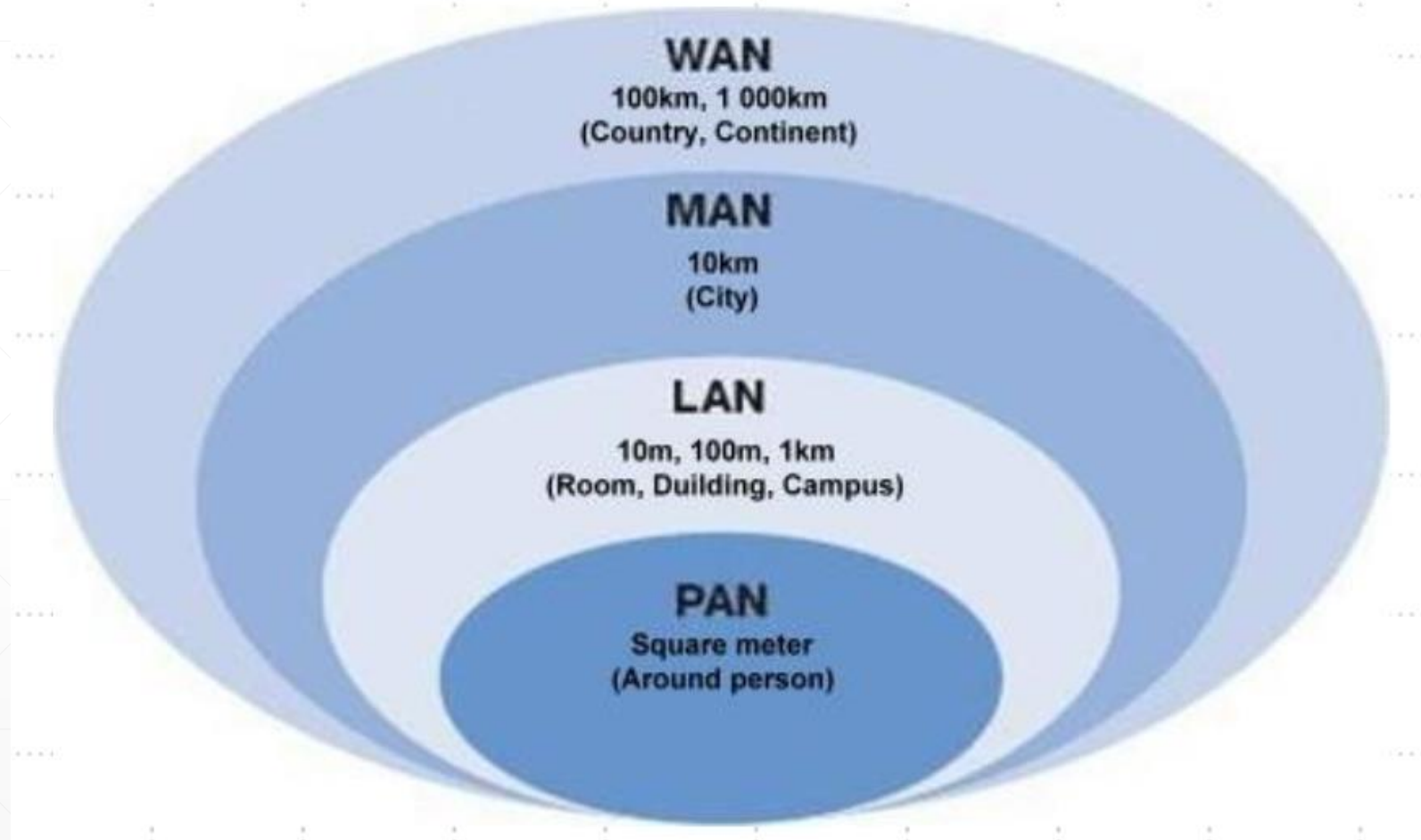


Physical Structures

Physical Topology

Aspect	Definition	Advantages	Disadvantages	
Mesh Topology	Every device has a dedicated link to every other device	- High robustness - Privacy/security - Easy fault detection	- Very costly - Complex cabling - Requires many I/O ports	
Star Topology	All devices connected to a central hub	- Easy to install/reconfigure - Robust (one link failure doesn't affect others) - Easy fault isolation	- If hub fails (central device), whole system fails - More cabling than bus	
Bus Topology	One long backbone cable shared by devices	- Easy installation - Less cabling	- Hard to reconfigure - Fault in backbone stops all communication - Signal degradation	
Ring Topology	Devices connected in a closed loop, each device (node) linked to 2 neighbors	- Easy installation - Simplified fault isolation - Reconfigurable	- Break in ring disables network - Unidirectional traffic (unless dual ring used)	
Hybrid Topology	Combination of two or more topologies	- Flexible - Can leverage strengths of multiple topologies	- Complex design - Expensive	

Categories of Networks

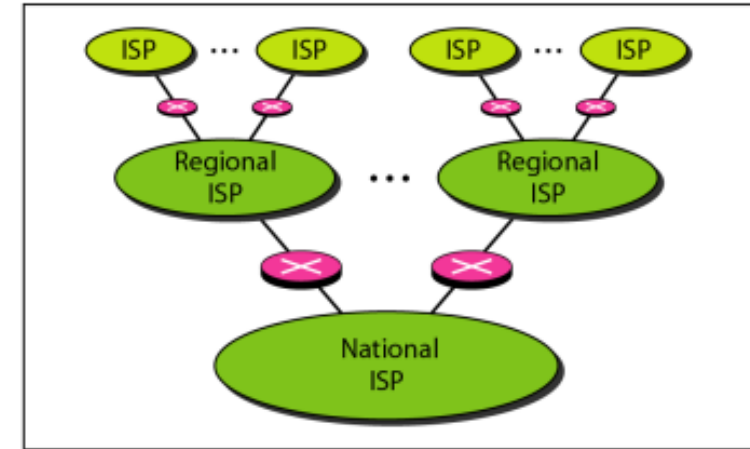


Categories of Networks

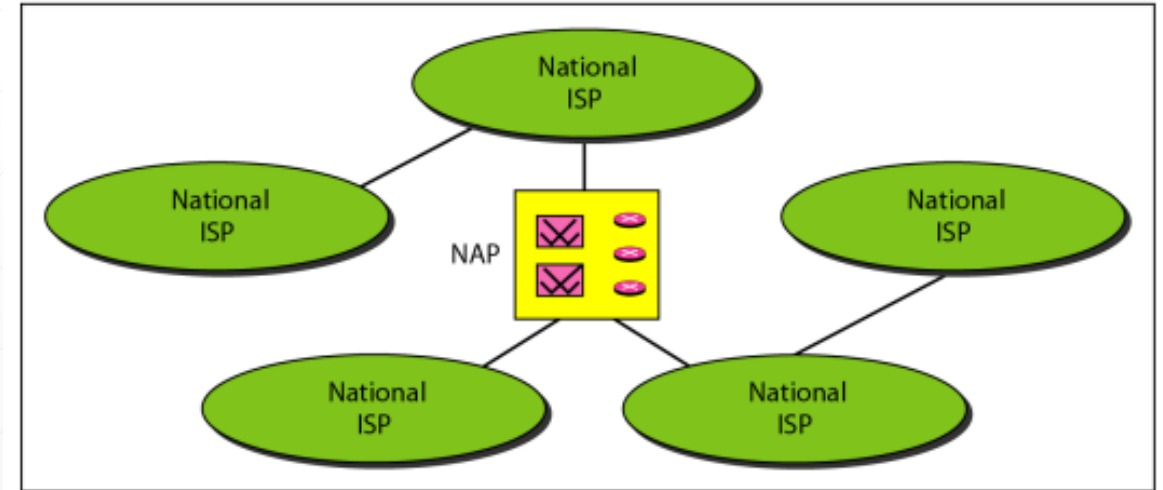
Type	Definition	Coverage / Size	Examples	Key Features
Local Area Network (LAN)	Privately owned network connecting devices in a small area (office, building, campus)	Up to a few km	Office PCs, printers, servers	- Resource sharing (HW, SW, data) - Topologies: bus, ring, star - Speeds: 100–1000 Mbps - Includes wireless LANs
Metropolitan Area Network (MAN)	Network larger than LAN but smaller than WAN, usually covering a city	Tens of km (town/city)	DSL, cable TV Internet	- High-speed connectivity - Serves organizations across a city - Acts as link between LANs and WANs
Wide Area Network (WAN)	Long-distance transmission over large geographic areas (country, continent, world)	Hundreds–thousands of km (global)	Internet backbone, ATM, Frame Relay	- Switched WAN (routers interconnected) - Point-to-point WAN (leased lines, DSL) - Includes wireless WANs
Internetwork (Internet)	Interconnection of LANs, MANs, and WANs into one system	Worldwide	Internet	- Rare to find isolated LAN/MAN/WAN - Provides global access & communication

Internet

- Today most end users who want Internet connection use the services of Internet service providers (ISP).
- There are :
 - International Service Providers
 - National Service Providers
 - Regional Service Providers
 - Local Service Providers



a. Structure of a national ISP



b. Interconnection of national ISPs

Protocols And Standards

- A protocol is a set of rules that lead data communications. An entity is anything capable of sending or receiving information. The key elements of a protocol are:
 - **Syntax:** refers to the structure or format of the data, meaning the order in which they are presented. Example:



- **Semantics:** refers to the meaning of each section of bits.
- **Timing:** refers to two characteristics: when data should be sent and how fast they can be sent